

E. Coli Plasmid Miniprep Protocol using Centrifugation

This protocol utilizes the Monarch Plasmid DNA Kit (NEB #T1110) kit, which allows for a reliable and quick method to purify up to 20 µg of high quality plasmid DNA and allows for 50 preps. This kit uses standard steps such as cell resuspension, alkaline lysis, and neutralization, and also includes color indicators at certain stages to help easily monitor when each step is complete.

Reagents

✓	Name	Quantity	Location/Notes
☐	Monarch Plasmid DNA Kit (NEB #T1110)		
☐	Antibiotic		
☐	Eppendorf tubes (2/ sample)		

Materials/Equipment

✓	Name	Quantity	Location/Notes
☐	Centrifuge		
☐			

Instructions

1. Night before: Incubate bacteria in 5mL of media overnight with appropriate concentration of antibiotics. Shake at ~225 rpm.
2. Pellet **3 ml** bacterial culture (not to exceed 15 OD units) by centrifugation for 30 seconds. Discard supernatant.
 - a. First put in ~1.5mL (no need to be too precise) of bacterial culture into an eppendorf tube and centrifuge for 30s.
 - b. Remove the supernatant into liquid waste.
 - c. Add in another 1.5mL of the bacterial culture into the same eppendorf tube and centrifuge for 30s. Discard supernatant.
3. Resuspend pellet in 200 μ L Plasmid Resuspension Buffer (B1)  (pink). Vortex or pipet to ensure cells are completely resuspended. There should be no visible clumps.
4. Lyse cells by adding 200 μ L Plasmid Lysis Buffer (B2)  (blue). Invert the tube immediately and gently 5–6 times until the solution is clear and viscous. **Do not vortex!** Incubate for one minute.
5. Neutralize the lysate by adding 400 μ L of Plasmid Neutralization Buffer (B3)  (yellow). **Gently** invert tube until color is uniformly yellow and a precipitate forms. **Do not vortex!** Incubate for 2 minutes.
6. Clarify the lysate by spinning for 5 minutes at 16,000 x g.
7. Carefully transfer supernatant to the spin column and centrifuge for 1 minute. Discard flow-through.
8. Re-insert column in the Monarch spin collection tube and add 200 μ L of Plasmid Wash Buffer 1. Plasmid Wash Buffer 1 removes RNA, protein and endotoxin. Centrifuge for 1 minute. Discarding the flow-through is optional.
9. Add 400 μ L of Plasmid Wash Buffer 2 and centrifuge for 1 minute.
10. Transfer column to a clean 1.5 ml microfuge tube. Use care to ensure that the tip of the column has not come into contact with the flow-through. If there is any doubt, re-spin the column for 1 minute before inserting it into the clean microfuge tube.
11. Add ≥ 30 μ L DNA Elution Buffer to the center of the matrix. **Do not touch the matrix with the pipette tip.** Wait for 1 minute, then spin for 1 minute to elute DNA.

Notes

- All centrifugation steps should be carried out at 16,000 x g (~13,000 RPM).
- If precipitate has formed in Lysis Buffer (B2), incubate at 30–37°C, inverting periodically to dissolve.
- Store Plasmid Neutralization Buffer (B3) at 4°C after opening, as it contains RNase A.

References

Monarch® Plasmid DNA Miniprep Kit Protocol (NEB #T1110)| New England Biolabs [Internet].
Available from:
<https://www.neb.com/en-ca/protocols/plasmid-miniprep-protocol-using-centrifugation-t1110>